CS 185/285 Assignment 5: Offline Reinforcement Learning — Report

All experiments ran locally on NVIDIA RTX A6000 GPUs with wandb in offline mode (`WANDB_MODE=offline`), so the training commands themselves match the assignment handout. Every run uses the default hyperparameters, seed 0 and 1M gradient steps; only the BC coefficient α is changed. The success rate is evaluated every 100K steps on 25 episodes, so a single evaluation has a resolution of 0.04 and a standard error of up to about 0.1. Differences of that size between two runs should be read as noise.

The raw runs of the α sweeps are kept in `exp_all/` (not part of `submit.zip`); the runs chosen for submission are copied to `exp/q{1,2,3}/{cube-single,antsoccer-arena}/`. The figures are produced by `uv run src/scripts/plot.py`.

Summary

Method	Task	Submitted α	Final success (1M steps)	Best success	Handout target	Met
SAC+BC	cube-single	300	1.00	1.00	> 0.75	yes
SAC+BC	antsoccer-arena	3	0.16	0.40	> 0.05	yes
IQL	cube-single	3	0.80	0.92	> 0.60	yes
IQL	antsoccer-arena	3	0.28	0.28	> 0.05	yes
FQL	cube-single	300	1.00	1.00	final > 0.80	yes
FQL	antsoccer-arena	10	0.28	0.52	best > 0.30	yes

Before the long runs, each implementation was checked on the optional debugging task `antmaze-medium-navigate-singletask-task1-v0` with $\alpha = 10$:

Method	Success at 100K steps	Success at 200K steps	Handout check
SAC+BC	0.92	0.92	> 0.80 at 100K: passed
IQL	0.72	0.84	> 0.60 at 200K: passed
FQL	0.88	0.84	> 0.80 at 200K: passed

Question 1: SAC+BC

Best runs.

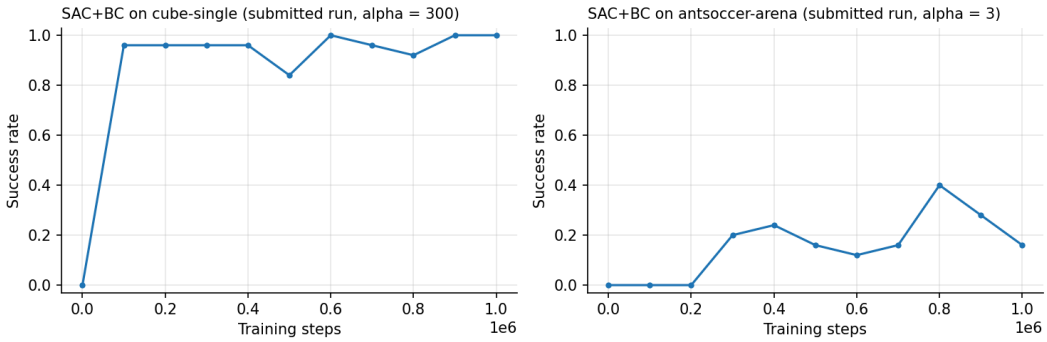


Figure 1: SAC+BC, submitted runs

α sweep. The handout asks for at least three values of α on cube-single; the right panel shows the same sweep on antsoccer-arena.

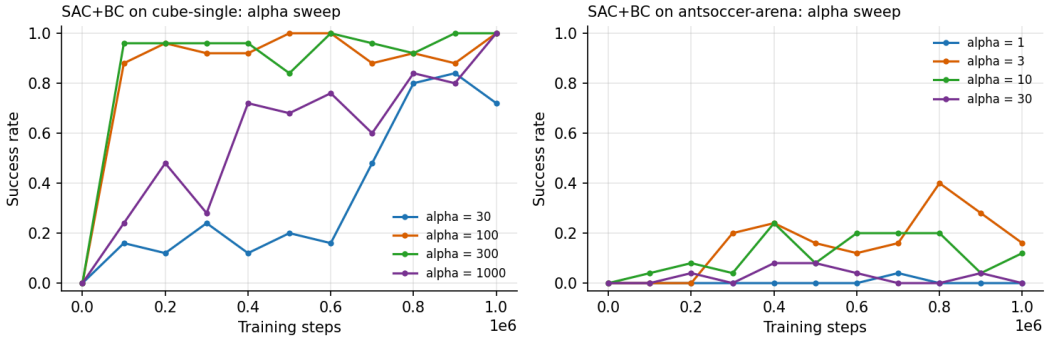


Figure 2: SAC+BC, alpha sweep

Task	α	Final success	Best success	Final action MSE	Final `q_min`
cube-single	30	0.72	0.84	0.047	−43.8
cube-single	100	1.00	1.00	0.040	−52.7
cube-single	300	1.00	1.00	0.038	−63.3
cube-single	1000	1.00	1.00	0.036	−83.4
antsoccer-arena	1	0.00	0.04	0.272	−100.1
antsoccer-arena	3	0.16	0.40	0.167	−102.1
antsoccer-arena	10	0.12	0.24	0.129	−99.7
antsoccer-arena	30	0.00	0.08	0.124	−99.8

On cube-single, $\alpha = 100$ and $\alpha = 300$ are at or above 0.84 from 100K steps onwards. $\alpha = 30$ regularizes too weakly: the policy drifts towards out-of-distribution actions with over-estimated values and stays below 0.25 for the first 600K steps. $\alpha = 1000$ is dominated by the BC term and improves slowly, although it also reaches 1.00 at the last evaluation. The MSE between policy and dataset actions decreases monotonically as α increases on both tasks, which is the sanity check suggested in the handout. `q_max` stays near 0 in all runs, and `q_min` is near −100 on antsoccer-arena and between −44 and −83 on cube-single, as expected for rewards in $\{-1, 0\}$ with $\gamma = 0.99$.

On antsoccer-arena, performance is low and noisy, as the handout anticipates. Only $\alpha = 3$ and $\alpha = 10$ are clearly non-zero; $\alpha = 1$ leaves the policy too far from the data (MSE 0.27) and $\alpha = 30$ is close to plain behavioral cloning.

Question 2: IQL

Best runs.

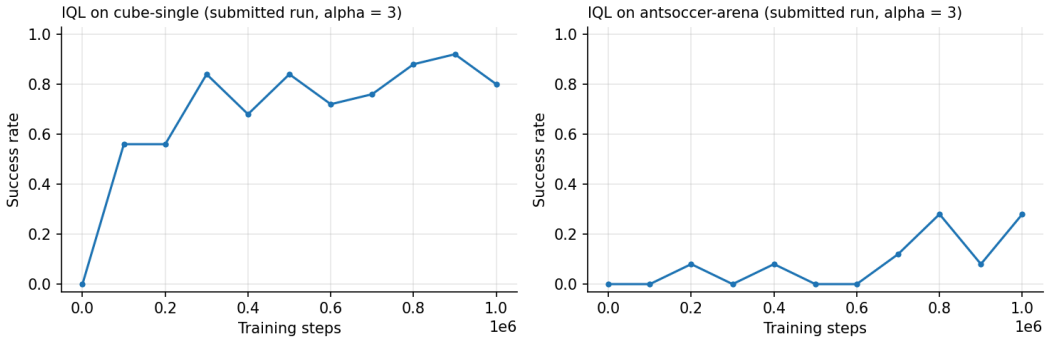


Figure 3: IQL, submitted runs

α sweep.

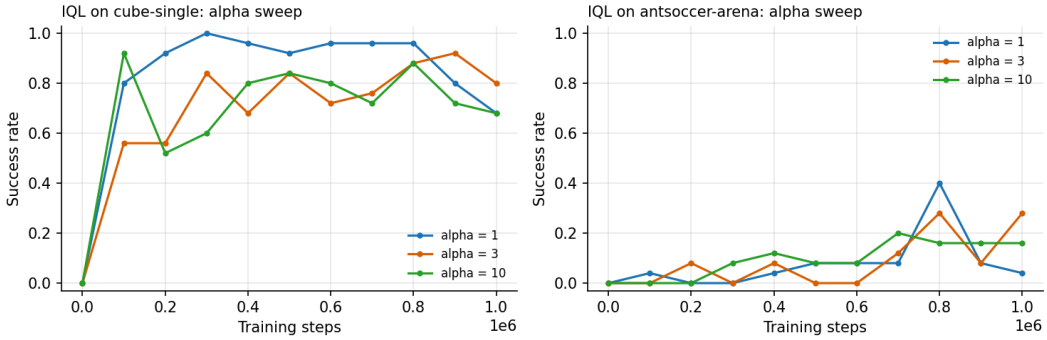


Figure 4: IQL, alpha sweep

Task	α	Final success	Best success	Mean of last 3 evals
cube-single	1	0.68	1.00	0.81
cube-single	3	0.80	0.92	0.87
cube-single	10	0.68	0.92	0.76
antsoccer-arena	1	0.04	0.40	0.17
antsoccer-arena	3	0.28	0.28	0.21
antsoccer-arena	10	0.16	0.20	0.16

In IQL the value functions are trained without the actor, so the Q and V curves are identical for all α with the same seed (final `q_min` = −54.4 on cube-single in all three runs); α only changes how sharply the advantage-weighted regression concentrates on high-advantage dataset actions.

SAC+BC versus IQL. (1) Best performance: SAC+BC is better on cube-single (1.00 at the end of training for $\alpha \geq 100$, versus 0.68–0.80 for IQL), because its actor directly follows the gradient of Q, while IQL's actor can only re-weight actions that already exist in the dataset. On antsoccer-arena both are low and within evaluation noise of each other (best final values 0.16 and 0.28). (2) Sensitivity to α: SAC+BC is much more sensitive. Its results on cube-single range from a run that stays below 0.25 for 600K steps (α = 30) to a run that is solved after 100K steps (α = 300), and on antsoccer-arena only the middle of the α range works at all. All three IQL runs on cube-single stay between 0.52 and 1.00 after the first 100K steps and end within 0.12 of each other, since IQL never evaluates the critic on out-of-distribution actions regardless of α.

Question 3: FQL

Best runs.

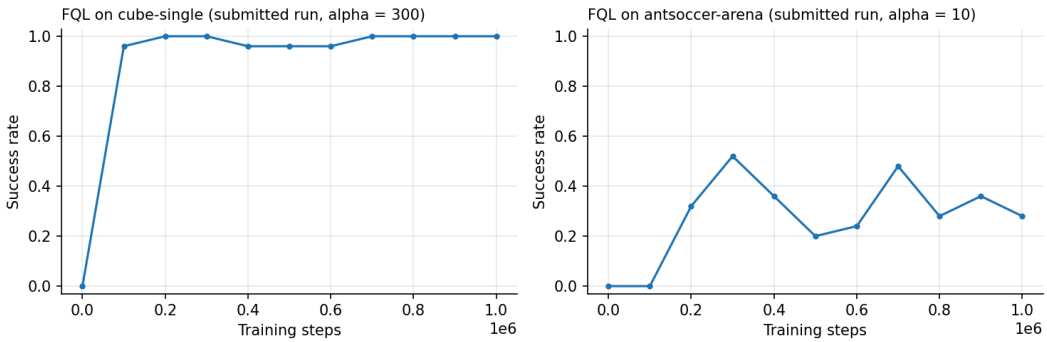


Figure 5: FQL, submitted runs

α sweep (not required, shown for completeness).

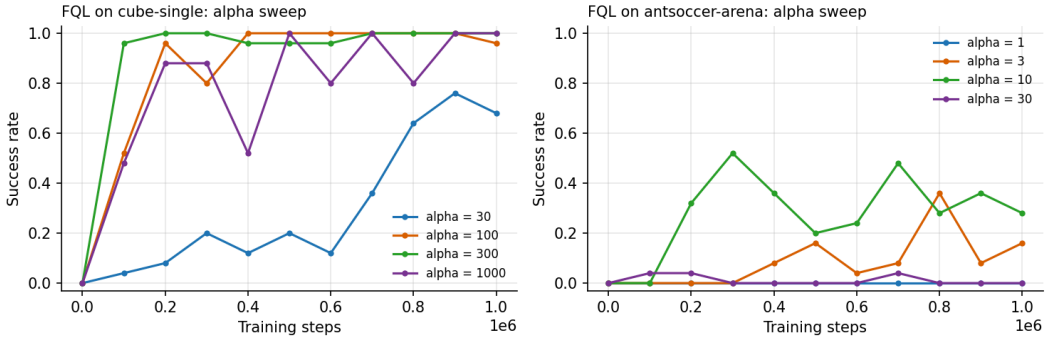


Figure 6: FQL, alpha sweep

Task	α	Final success	Best success	Final one-step action MSE
cube-single	30	0.68	0.76	0.032
cube-single	100	0.96	1.00	0.024
cube-single	300	1.00	1.00	0.027
cube-single	1000	1.00	1.00	0.026
antsoccer-arena	1	0.00	0.00	0.318
antsoccer-arena	3	0.16	0.36	0.228
antsoccer-arena	10	0.28	0.52	0.198
antsoccer-arena	30	0.00	0.04	0.191

FQL with $\alpha = 300$ is at or above 0.96 on cube-single at every evaluation from 100K steps on and ends at 1.00. On antsoccer-arena, $\alpha = 10$ is the only setting that works well: it reaches 0.52 at 300K steps and stays between 0.20 and 0.48 afterwards, which is the highest antsoccer-arena result among the three methods. The expressive flow policy can represent the multi-modal behavior data of this task, which a unimodal Gaussian actor cannot. As with SAC+BC, the result is sensitive to α on both sides: $\alpha = 1$ and $\alpha = 30$ both give essentially zero success on antsoccer-arena.

Commands

```
export WANDB_MODE=offline MUJOCO_GL=egl

# Question 1 (SAC+BC)
for a in 30 100 300 1000; do
  uv run src/scripts/run.py --run_group=q1 --base_config=sacbc --env_name=cube-single-
play-singletask-task1-v0 --seed=0 --alpha=$a
done
for a in 1 3 10 30; do
  uv run src/scripts/run.py --run_group=q1 --base_config=sacbc --env_name=antsoccer-
arena-navigate-singletask-task1-v0 --seed=0 --alpha=$a
done

# Question 2 (IQL)
for a in 1 3 10; do
  uv run src/scripts/run.py --run_group=q2 --base_config=iql --env_name=cube-single-
play-singletask-task1-v0 --seed=0 --alpha=$a
  uv run src/scripts/run.py --run_group=q2 --base_config=iql --env_name=antsoccer-arena-
navigate-singletask-task1-v0 --seed=0 --alpha=$a
done

# Question 3 (FQL)
for a in 30 100 300 1000; do
  uv run src/scripts/run.py --run_group=q3 --base_config=fql --env_name=cube-single-
```

```
play-singletask-task1-v0 --seed=0 --alpha=$a
done
for a in 1 3 10 30; do
  uv run src/scripts/run.py --run_group=q3 --base_config=fql --env_name=antsoccer-arena-
navigate-singletask-task1-v0 --seed=0 --alpha=$a
done

# Figures
uv run src/scripts/plot.py
```